Semantic-Aware Loss For Facial Landmark Detection

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Abstract

Recent research of face alignment has noticed the problem of semantic ambiguity of facial landmarks in annotations due to the unclear definition of the problem, which might be an important factor of its performance degradation. In this paper, we present a semantic-aware loss which tries to minimize the normal distances between the predicted points and the boundary corresponding ground truth points by changing the ground truth points adaptive during training. Specially by using the state-of-art method Convolutional Neural Network, we propose a novel framework Semantic Regression Network which combine CNN and our semantic-aware loss. In order to minimize normal distance, we propose two different methods, an analytical method and a sampling method, to make our loss differentiable in training neural network. We also found that the two methods can be used in traditional algorithms, not just deep learning models. Furthermore, inspired by the previous work Wing Loss, we try to give a quantitative analysis for better explaining why our method work, rather than just relying on intuition. At last, by rethinking the evaluation metric Normalized Mean Error (NME), we propose a new metric that makes a clearer definition to the problem for evaluating the performance of face alignment in practice. Experimental results on AFLW and 300W demonstrate the effectiveness of our method and prove the superiority of over the state-of-the-art approaches.

1. Introduction

Face alignment, or facial landmark location, is an active computer vision domain which aims to find a set of predefined 2D/3D key points in a face image. These key points usually have a semantic meaning e.g. nose tip or eye center, which provide a lot of geometric information for subsequent task, such face identification/verification, pose estimation or 3D reconstruction.

Despite recent development such as deep learning methods has achieved great progress, the performance of facial landmarks location is still suffered from ”semantic ambiguity”. The ambiguity is caused by the unclear definition of the problem. For example, the definition of the key points on the face contour have a clear geometric meaning to present the curve of the face contour boundary, but leads to the ambiguity of the precise location for each point. Because any ”small semantic movement” [9] on the boundary is reasonable, and the ground truth of facial landmark location is annotated by annotators which is another important source of ambiguity. LAB [17] and GHCU [9] also have realized such problem, and try to solve this by a boundary-aware framework and a global heatmap correction unit. In practice, such ambiguity is inevitable for annotators which is an important reason of the performance degradation. Clearly, even for state-of-art methods such CNN, this ambiguity will lead the model training trapped into local minimum.

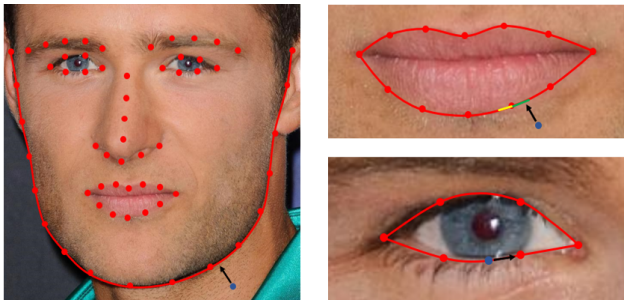
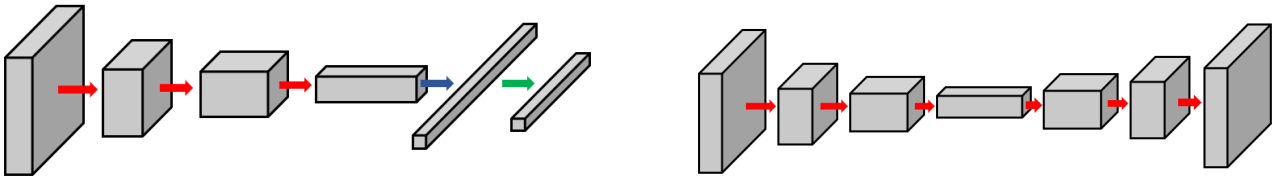


Figure 1. Red dots and blue dots present ground truth and predicted key points, and red line present the face contour (semantic boundary). (a) Left. Non-semantic movement, that is, we want to optimize our model that the predicted(blue) dot falls as close as possible to on the semantic boundary(red). (b) Non-semantic movement, yellow and green line depict a reasonable range of optimization direction of the blue dot. Or in other words, points on yellow and green line are semantically equivalent to the red point (the intersection of the two segments). (c) Semantic movement, which means the optimization direction can be ignored during training.

In this paper, we propose a novel semantic-aware loss to eliminate the semantic ambiguity of ground truth points. Specially for CNN, we present Semantic Regression Network (SRN) and two different methods that act as an auxiliary process for adaptive changing the location of ground



(a) regression (b) heatmap

Figure 2. two classical architecture of network for facial landmark localization

truth each point. Figure 1(a) shows that our semantic-aware loss has a clear geometric meaning which tries to minimum the distance between each pair of predicted point and its corresponding semantic boundary. In practice, the semantic boundary of each key point is usually inaccessible due to the lack of analytic expression of the curve (e.g. the face contour), so we change our target to minimum the distance between the key point and two semantic segments which the ground truth point relies on. Figure 1(b) depicts the two semantic segments, they present a reasonable range of "semantic movement". The two semantic segments are small part of the two segments which are the connections between before and after point and current point. As a result, the loss of minimum the distance between the key point and two semantic segments act as a proxy loss function of minimization the distance. Specially for CNN, for minimization the distance, we present an analytical method and a sampling method, both of which can make our Semantic Regression Network train in end-to-end manner. The contributions of our work include:

- Presenting a semantic-aware loss which directly tries to minimum the distance between each pair of predicted point and its corresponding semantic boundary. Specially, combining with convolution neural network, two methods is presented to make the network be trained in an end-to-end manner.
- quantitative analysis is given to for better understanding of our work, and notice that our method and Wing Loss [17] is similar in the actual effect of gradient update, although the starting point is completely different.
- A new evaluation metric named SNME (Semantic Normalized Mean Error) is proposed which contributes to make the definition of the problem clearer for better evaluating the performance of face alignment in actual.

2. Related Work

2.1. Architecture

In the literature, we can simply divide the methods of face alignment into two categories: classic methods (ASM, ESR, AAM, CLM) and deep learning methods. Our approach Semantic Regression Network is based on deep learning methods. Referring the view of architecture from literature [17], we also divide methods into two groups, coordinate regression model and heatmap regression model.

Coordinate regression model usually directly learns the map of face images to facial landmarks coordinate with Deep Convolutional Neural Networks. The architecture of these methods is straightforward which use Convolutional Neural Network to extract features and output the 2D coordinate of facial landmarks in the final layer which presents by a vector. The pipeline is very simply which only consists of convolution, pooling and fully connected layer **making it widely used in practice**. Based on this simply framework, a lot of efforts have been made to try to improve the performance. These works include multi-task [21] and cascade network [15, 22, 10] etc. Figure 2(a) depicts a simply pipeline of this architecture.

Heatmap regression model uses the fully convolutional network to generate a likelihood heatmap for each facial landmark which recently have achieved state-of-art performance in face alignment. Some typical methods include [17, 5, 1, 6]. The architecture is designed based on Stack Hourglass Network which is originally used for human pose estimation [11]. Its architecture usually consists of an encoder and a decoder which both are composed of several convolution, pooling layers. An important contribution of this method is 3D face alignment [2] which, to our best knowledge, is hard for coordinate regression model. Due to its complex architecture, the cost of the parameters and calculations are huge.

2.2. Semantic ambiguity

As mentioned above, the semantic ambiguity is caused by an unclear definition of some facial landmarks. Referring the view of the ambiguity from literature [9], it can have potentially influence: (1) the accuracy of the annotations, (2) the convergence of model training and (3) the evaluation metrics of model performance.

For (1), we did a small experiment to verify our ideas about annotations. We hired a group of volunteers to make annotations of facial landmarks and collected the results. Then, we use the heatmap to visual the annotations to help use understand the problem of semantic ambiguity of an-

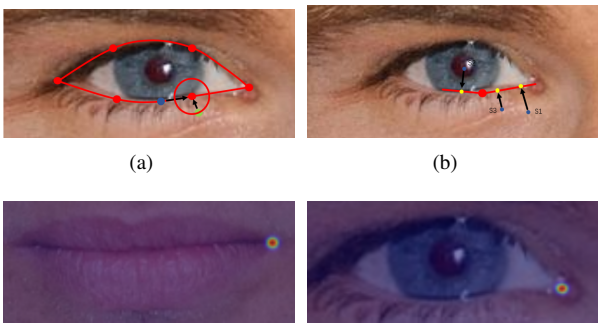


Figure 3. (a) from the perspective of evaluation, the green dot should get better results on NME, but the fact is opposite. All NMEs of the points on red circle around the ground truth are equal, this result is counterintuitive. (b) Red dot presents the original ground truth, red lines are the corresponding semantic boundary(segments). Blue dots are the predicted points at different steps, yellow dots are the corresponding adaptive ground truths during training at different steps, and black arrows indicate their optimization directions. In order to express our meaning more clearly, we exaggerate the length of the semantic segments. (c) visualization of right corner of mouth. (d) visualization of right corner of the eye. The heatmap shows the probability distribution of pixels marked as key points.

notations. Figure 3 (c) and (d) give two samples about the this issue. For (2), the ambiguity of ground truth key points affects the gradient which makes the model converge to a worse local optimum. This result is expressed that the prediction of some points with less semantic ambiguity is worse than these with more semantic ambiguity. For (3), its negative effect is obvious because a point which relying on semantic boundary is better than the one which not relying on, but the later maybe have a better performance on the metric NME. Figure 3 (a) show a simply case on an eye.

In actually, some works have been made recently to try to overcome this issue. LAB [17] tries to train an extend facial boundary heatmap estimator and incorporates it into the landmark localization network. The facial boundaries provide an explicit geometric structure to overcome the semantic ambiguity of ground truths. But the computation is very expensive due to its architecture design. GHCU [9] model the 'real' ground truth as a latent variable to optimize, and the optimized 'real' ground truth then supervised the landmark detection network training. This method is a typical process of EM algorithm that the alternative optimization process can be seen as the two steps of EM. Although their motivation is to solve the problem of semantic ambiguity of facial landmarks, but the hypothesis behind is still suffered the problem of semantic equivalence, shown in figure 1(b). Also, due to usage of probabilistic model, GHCU suffers the same problems as LAB in computation.

3. Semantic-aware loss

3.1. CNN-based regression network for facial landmark localization

Figure 2(a) has depicted a simply pipeline and architecture of CNN-based regression network. Here we can define its mathematical expression, it means that given a set of input images $x \in R^{H\times W\times 3}$and labels $y \in R^{2\times T}$, we can define the nonlinear mapping (which presents the process of convolution neural network):

$$\Phi : x \rightarrow y \tag{1}$$

the target (or loss function) is to find that Φ minimums:

$$L = \sum_{n=1}^{N}\sum_{t=1}^{T} F(\hat{y}_t^n, y_t^n) \tag{2}$$

Where y_t^n is the prediction for y^n, and F is a pre-defined function (such as L1 or L2) that measures the difference between the predicted points and its corresponding ground truth. For optimizing the loss function, algorithms such as Ada or SGD can be used.

3.2. Semantic-aware loss

As mentioned above, our target is to minimum the distance between each point and its "semantic boundary curve" of corresponding ground truth. The "semantic boundary curve" is part of the whole boundary, and the points on the semantic boundary can maintain the semantics of the point. Or in other words, semantic boundary of each point gives a reasonable range of semantic movements. But due to the lack of the analytical expression, it is hard to calculate the distance between each point and its "semantic boundary curve". So, we need to find a proxy function to calculate the distance. **The most direct way is to replace the curve(boundary) by the two segments which are connecting front and back point (the motivation behind is the clear geometric meaning of each facial landmark)**. It is worth to notice that not all points on the two segments can be considered as ground truth, and only a small part of them are reasonable. As a result, we replace each ground truth with the two segments and make it as target to optimize. We can obtain the two segments by intercepting two semantic endpoints. Then we can define our new loss function as:

$$L_{SAL} = \sum_{n=1}^{N}\sum_{t=1}^{T} \min[G(\hat{y}_t^n, s_t^n), G(\hat{y}_t^n, e_t^n)] \tag{3}$$

where s_t^n and e_t^n represents the corresponding two semantic segments of each ground truth point y_t^n, and G is the function to calculate the distances. Figure 1(b) shows an example on a mouth. But there is a problem that function

G is not differentiable. Specially, to solve the problem and make the network that can be trained in end-to-end manner, we propose two methods: analytical method and sampling method.

3.3. Analytical method of semantic-aware loss

Analytical method is strategy to adaptively change ground truth points during training. Figure 3 (b) shows a process that how to change ground truth points adaptively. At each iteration, we calculate the closest point to the predicted point which relies on s_t^n or e_t^n, and make this point as the ground truth. The new adaptive ground truth has a clear geometric meaning that it is the closest semantic point (which is corresponding to the predicted point). It will adaptively change the optimization directions according to the current predicted points. And this adaptive strategy has a hidden effect, reducing the overall value of loss which will reduce the overfitting. Table 1 gives whole pipeline of this method.

3.4. Sampling method of semantic-aware loss

It is obvious that analytical method is time consuming because of the process of calculating new ground truths at each step. As a result, we proposal the sampling method to overcome this issue. Figure 4 show an example of this sampling. The key idea is to replace the two semantic segments with several semantic points which are sampled uniformly on the two. These sampling points (or semantic points) are presentations of two semantic segments (or semantic boundary). Details can be followed in Appendix. The number of the sampling points is a hyperparameter. And it is obvious that more points will represents the semantic better. Then the loss function will be defined as:

$$L_{SAL} = \sum_{n=1}^{N} \sum_{t=1}^{T} H(\hat{y}_t^n, y_{t0}^n, y_{t1}^n, ..., y_{tm}^n) \tag{4}$$

where T is the number of sampling points, and y_{t0}^n is the original ground truth y_t^n, and $y_{tm}^n (i = 1, ..m)$ represents the corresponding sampling points of y_t^n, H is the minimum function defined as:

$$H(\hat{y}_t^n, y_{t0}^n, y_{t1}^n, ..., y_{tm}^n) = \min_{i=1,...,m} F(\hat{y}_t^n, y_{ti}^n) \tag{5}$$

Actually, the sampling method can be view as an exchanging strategy between time and space. The most time-consuming process (calculating adaptive labels during training) has already been finished before training (i.e. sampling process). The only difference is that the sampling point which is ultimately used for the loss calculation may not be a vertical point. This will result a greater loss value during training (comparing with the vertical point). But with the increasing number of sampling points, the difference of loss value can be ignored.

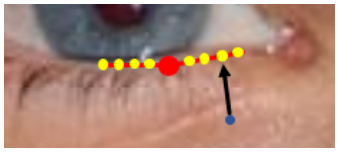


Figure 4. Red dot is the original ground truth, yellow dots are the sampling points, blue dot is the predicted point. Black arrow indicates their optimization direction (which has the minimum distance between predicted point and all the sampling points).

4. Experiments

4.1. Datasets

We conduct evaluation on three challenging datasets including 300W [14], COFW [3], WFLW [17].

300W dataset is a collection of multiple face datasets, including LFPW, HELEN, AFW, XM2VTS that all of them have 68 landmarks. To perform the evaluation on this dataset, we follow the protocol used in [13]. The protocol uses the full set of AFW and the training subsets of LFPW and HELEN as the training set, which contains 3148 training samples. The test set have 689 samples which consists of the test subsets of LFPW and HELEN, as well as the whole of IBUG. All the test samples are further divided into common and challenging subsets.

COFW dataset consists of 1345 samples for training and 507 for testing which are all occluded to different degrees. The COFW dataset originally has 29 landmarks. We also use the test set which has been re-annotated by [7] with 68 landmarks annotation scheme to allow easy comparison to previous methods.

WFLW dataset was introduced by [17], the full name is Wider Facial Landmarks in-the-wild. It contains 10000 samples (8000 for training and 2000 for testing) with 98 annotated landmarks. Also, WFLW contains more rich annotations such as occlusion, pose, make-up, illumination, blur and expression for future research of face alignment.

4.2. Evaluation metrics

To evaluate our facial landmark localization algorithm, we adopted the widely used Normalized Mean Error (NME). Specially, we also use the inter-pupil distance as the normalization term, which is different from the standard 300W protocol that uses the outer eye corner distance. Meanwhile, according to the previous analysis about the semantic boundary, we also use Semantic Normalized Mean Error (SNME) as another metric to better evaluate the performance of face alignment methods. **NME** calculate the mean distances between the predicted points and ground-truth points. That is,

$$NME = \frac{1}{T} \sum_{t=1}^{T} \|\hat{y}_t^n, y_t^n\|_2 \tag{6}$$

Where T is the number of landmarks, and usually we calculate across the entire test set. **SNME** is different from the NME which calculate the mean distances between the predicted points and the corresponding semantic boundaries of both sides of the ground-truth points. That is,

$$SNME = \frac{1}{T}\sum_{t=1}^{T} G(\hat{y}_t^n, y_t^n) \tag{7}$$

where $G(\hat{y}_t^n, y_t^n)$ is the distance between the predicted points and the corresponding semantic boundaries (for better understanding, please refer to Figure 3 and Figure 4).

4.3. Implement details

For fair comparison between different methods, we use the same training configuration which includes input size, backbone, optimization method, number of iterations, learning rate, etc. Specially in our experiment, we used MobiletNetV3 as backbone, and Adam as optimizer. Further in details, we set momentum to 0.9 and weight decay to 10e-4 in optimization. Also, input image size is 256, batch size of each iterations is 128 and over all training epoch is 1200. Learning rate decreases linearly from 10e-3 at epoch 1 to 10e-6 at epoch 1000 and keeps 10e-6 at the last 200 epochs. On the condition of data augmentation, we randomly rotated each training image between [-20, +20] degrees and randomly pad and crop [-0.15, +0.15] of the input image size. For Wing Loss, we select the best hyper-parameter in [17], that is, ω is 10 and ϵ is 2. For the sampling method of semantic-aware loss, we set 9 to the number of sampling points for comparison. We implement 6 different loss function in pytorch, and train all of them on a single server(Ubuntu 18.4 with Intel i7 CPU and Nvidia TITAN RTX GPU).

4.4. Comparison with other approaches

4.4.1 Evaluation on 300W

For the dataset 300W We use NME and SNME for better verifying our method. For NME, the results are shown in the Table 1. Our method achieve a very comparable result on the 300W dataset in accuracy.

For a fairer comparison on SNME, we just use three of them for reference, and the results are shown in Table 2. It is clear, that our method achieve the best on the three subset.

4.4.2 Evaluation on WFLW

For comprehensively evaluating different methods, WFLW dataset is proposed by [17] which diversifies in head pose, expression, illumination makeup, occlusion and blur. Due to the limitations of regression methods, the landmark cannot be expressed under the extreme head pose condition. As a result, we remove the extreme head pose faces in both train set and test set. Also, for cross-dataset evaluation, we just sample 68 points from its annotated 98 points according to the previous definition in 300W dataset. In this section, we just compare several losses based regression method which is more meaningful in practice. We train our model on WFLW-Trainset and test on WFLW-Testset. Also, we do cross-dataset evaluation on Trainset and Testset of 300W

Table 1. Add Evaluation of NME on 300W dataset

Method	Common	Challenge	Full
RCPR[3]	6.18	17.26	8.35
CFAN[20]	5.5	16.78	7.69
ESR[4]	5.28	17	7.58
SDM[19]	5.6	15.4	7.52
LBF[12]	4.95	11.98	6.32
CFSS[23]	4.73	9.98	5.76
3DDFA[24]	6.15	10.59	7.01
TCDCN[22]	4.8	8.6	5.54
MDM[16]	4.83	10.14	5.88
RAR[18]	4.12	8.35	4.94
DeFA[8]	5.37	9.38	6.1
HGs[9]	4.43	7.56	5.04
LAB[17]	3.42	6.98	4.12
L1	3.31	6.44	3.94
SmoothL1	3.37	**6.26**	3.96
Wing[17]	**3.29**	6.57	3.96
SAL	3.32	6.34	**3.94**

Table 2. Evaluation of SNME on 300W dataset

Method	Common	Challenge	Full
L1	1.67	3.97	2.14
SmoothL1	1.7	3.76	2.12
Wing	1.67	4.06	2.15
SAL	**1.42**	**3.64**	**1.87**

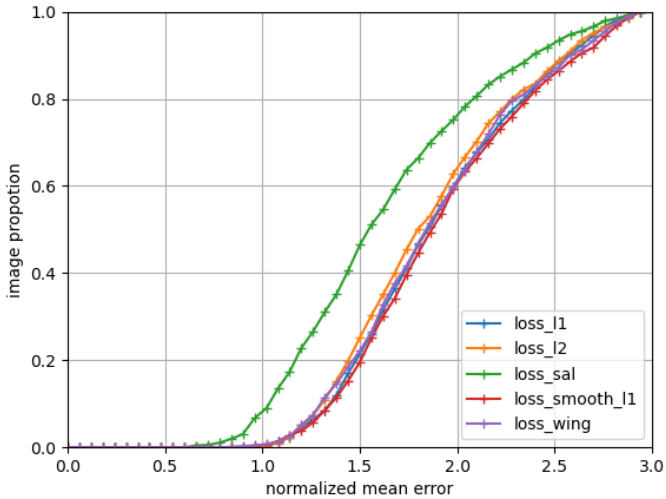


Figure 5. CED curves of different methods on dataset 300W

dataset, COFW dataset (for COFW we use both trainset and testset for test).

Table 3. Evaluation of SNME on WFLW, 300W and COFW.

Method	WFLW	COFW	300W			
			Train	Comm	Challe	Full
L1	5.39	3.39	1.39	1.67	4.03	2.15
L2	5.6	3.39	1.48	1.81	4.18	2.29
SmoothL1	5.48	2.42	1.4	1.67	3.71	2.08
Wing	5.37	2.4	1.34	1.68	3.85	2.12
SAL	**5.21**	**1.9**	**1.27**	**1.47**	**3.78**	**1.94**

As the result shown in Table 3, SAL out-perform all other regression method in all three dataset under the metric of SNME. Meanwhile, Figure 5 shows the CED curves of different methods, and our method against all others on 300W-trainset.

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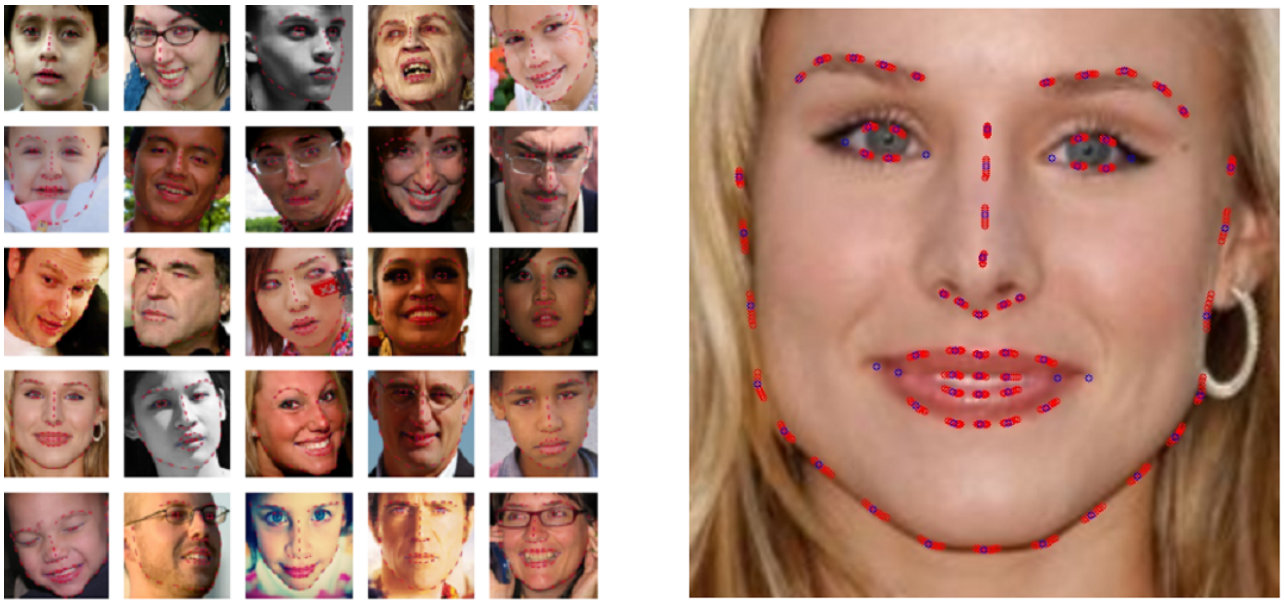

(a) (b)

Figure 6. The left image grid shows our sampling points for SAL, the right image shows some detail.

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5. Appendix

Figure 6 give some sampling images to show the detail of semantic points. For detail in right points, the blue points represent the original ground-truth which is manually annotated, and the red points represent the semantic points which is corresponding to its ground-truth points. Every ground-truth point can sample several semantic points (the number of semantic points is a hyperparameter, our experiment uses 9) on the semantic boundary in a limited distance. Specially, some of the landmarks do not have semantic points (in other words, the coordinates of all its semantic points are the same), such as the inner and outer corners of both eyes. These ground-truth points are marked with a single blue point without any red points around.